

Figure 5. Method novelty audit: what existing multimodal integration methods already do

MVAE (PoE) (2018)	yes	yes	yes	yes	yes	no	no	no	no
MMVAE (2019)	yes	yes	no	no	yes	no	no	no	no
MoPoE-VAE (2021)	yes	yes	yes	yes	yes	no	no	no	no
gPoE-VAE (2022)	yes	yes	yes	yes	yes	no	no	no	no
gPoE-normVAE (2023)	yes	yes	yes	no	yes	no	no	no	no
COLD Fusion (2023)	yes	yes	yes	yes	no	yes	no	no	no
totalVI (2021)	no	no	no	no	yes	yes	no	no	no
MultiVI (2023)	yes	yes	no	yes	yes	no	yes	no	no
Cobolt (2021)	yes	yes	yes	yes	yes	no	yes	no	no
scMM (2021)	yes	yes	no	no	yes	no	no	no	no
scMaui (2024)	yes	yes	yes	yes	yes	no	no	no	no
UniVI (2025)	yes	yes	yes	yes	yes	no	yes	no	no
MOFA+ (2020)	no	no	no	no	yes	no	no	no	no
	modality specific encoders	per modality posterior variance	precision weighted fusion	cell specific weighting	missing modality support	calibration evaluated	domain shift evaluated	selective prediction evaluated	source composition perturbation